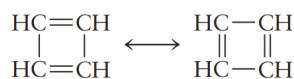
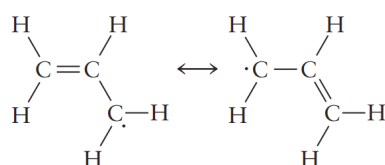


- Describe the bonding in the bent molecule OF_2 . Predict its energy level diagram and electron configuration.
- Discuss the nature of the bonding in the nitrate ion (NO_3^-). Draw the possible Lewis resonance diagrams for this ion. Use the VSEPR theory to determine the steric number, the hybridization of the central N atom, and the geometry of the ion. Show how the use of resonance structures can be avoided by introducing a de-localized π MO. What bond order is predicted by the MO model for the N-O bonds in the nitrate ion?
- For each of the following molecules, construct the π MOs from the $2p_z$ atomic orbitals perpendicular to the plane of the carbon atoms.

(a) Cyclobutadiene



(b) Allyl radical



Indicate which, if any, of these orbitals have identical energies from symmetry considerations. Show the number of electrons occupying each π MO in the ground state, and indicate whether either or both of the molecules are paramagnetic. Assume that the C atoms in the allyl radical are all sp^2 hybridized.

- Write an equation involving structural formulas for the catalytic cracking of 2,2,3,4,5,5-hexamethylhexane. Assume that the cracking occurs between carbon atoms 3 and 4.
 - Draw and name one other isomer of the alkene.
- Write structural formulas for the following:
 - 2,3-Dimethyl-1-cyclobutene
 - 2-Methyl-2-butene
 - 2-Methyl-1,3-butadiene
 - 2,3-Dimethyl-3-ethylhexane
 - 4,5-Diethyloctane
 - Cyclooctene
 - Propadiene
 - 2-pentyne